

Pre-Chamber Ignition of Ammonia and Hydrogen Fuels for Marine Engines

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Introduction

Global greenhouse emissions have been a matter of concern for many decades with the largest contribution coming from the combustion product of fossil fuels. Among the major sectors where the emissions come from, almost three-quarter come from energy consumption, especially transportation, which account for 16.2% of the global greenhouse gas emissions [1]. Greenhouse gas emissions from transportation primarily come from combustion of fossil fuel used in cars, trucks, ships, trains, and airplanes. Over 94% of the fuel used for transportation is petroleum based, which includes primarily gasoline and diesel [2]. Several experimental works have reported the feasibility of using ammonia as an alternative for internal combustion engines (ICE). Nevertheless, due to the properties of ammonia such as high auto ignition temperature and low laminar burning velocity, there are challenges in maintaining sufficient combustion of ammonia [3]. However, it is possible to achieve satisfactory combustion by using some type of combustion promoter. Common fuels used in ICEs such as gasoline and diesel, as well as alternative fuels like hydrogen, biodiesel, methanol can be used as combustion promoters [4]. To ensure efficient and complete combustion, hydrogen is considered the best promoter because it has characteristics that are opposite to and complement those of ammonia. Hydrogen possesses a rapid combustion rate, low ignition energy, and an extensive range of flammability [5, 6].

Although, the combination of ammonia and hydrogen results in a stable combustion, significant emissions of NO_x and low combustion efficiency have been observed [7]. The implementation of a pre-chamber is a promising method to enhance the combustion quality, as demonstrated in conventional engines [8]. Simulation results show that a SI engine with a pre-chamber installed operating under lean conditions provides higher indicated thermal efficiency compared to the normal SI mode [9]. Adding hydrogen to a single cylinder engine with a pre-chamber increased the efficiency, gross work, and indicated mean effective pressure, while also improving combustion quality and reducing emissions by 20% and 15% [10]. While the pre-chamber has shown promising results, it is a relatively new concept for ammonia that requires further investigation to fully understand its potential benefits and limitations.

The study will focus on the ignition processes when an ammonia and hydrogen mixture is burnt inside an internal combustion. The main objective of the work will be to develop a suitable pre chamber which achieve lean burn combustion of ammonia and low emissions of nitrogen oxides and nitrous oxide. To meet the objectives a mixture of experimental and numerical work will be conducted. The project will be used to develop new engines and support new numerical models describing combustion of the fuel mixture.

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Methodology

To fully understand the combustion of ammonia and hydrogen mixtures in marine engines, this study will first develop a detailed computational fluid dynamics model to fundamentally understand in-cylinder combustion processes with different energy share ratios in a pre-chamber ignited marine engine. The effectiveness of hydrogen in improving combustion quality with ammonia fuelling will be assessed over the in-cylinder pressure-temperature domain using chemical kinetics modelling. Furthermore, for the experimental campaign, active pre-chambers with various geometry designs will be developed to investigate their effects on combustion and performance in ICEs.

Numerical Methodology

The LOGE software package will be used as a reaction kinetics solver to perform simulations such as ignition delay and laminar flame speed calculation. A closed, constant volume homogeneous 0-D batch reactor with adiabatic and chemically inert walls is used to calculate the ignition delay time of the air-fuel mixture, representing the in-cylinder trapped charge under engine pressure and temperature conditions. Additionally, the combustion and chemical kinetics simulations will be performed to predict NOx emission levels. Furthermore, a multidimensional, computational fluid dynamics simulation model using Converge CFD Software will be developed to predict the spray and combustion characteristics of hydrogen and ammonia mixtures inside a pre-chamber and in-cylinder under various conditions.

Experimental Activity

For all tests, the pre-chamber with various geometry and nozzle designs will be employed as an ignition system. Figure 1. illustrates the design of the pre-chamber with a 6-holes coaxial arrangement of nozzle and its cross section. It has been reported that the small pre-chamber concept (volume is usually less than 5% of the engine clearance volume) proposed by Gussak [11] is one of the most investigated concepts in recent years. Inspired by Hydrogen Assisted Jet Ignition system [12] a new pre-chamber unit was designed to be installed into a constant volume combustion chamber (CVCC) and optical accessible compression ignition chamber (OACIC) without significant modifications. In order to capture the turbulent jet from the pre-chamber, high speed imaging will be utilised to capture the spray images as the jet enters the main chamber.

To capture full images of the jet and spray angle and characteristics of the pre-chamber, a CVCC with optical imaging techniques will be utilised. The chamber consisting of an air supplying system to introduce a mixture of O_2 , N_2 and CO_2 ; these gases will be mixed using a mixing fan to form a homogeneous mixture. A Bosch gasoline direct injection (HDEV5) will supply hydrogen, hydrogen/nitrogen or hydrogen/ammonia mixtures into the pre-chamber. A spark-plug type pressure transducer will be installed in the chamber for combustion measurement and pressure trace recording. Figure 2. shows a 3D CAD drawing of a CVCC with a pre-chamber on the top and the nozzle location when viewing from the front of the chamber.

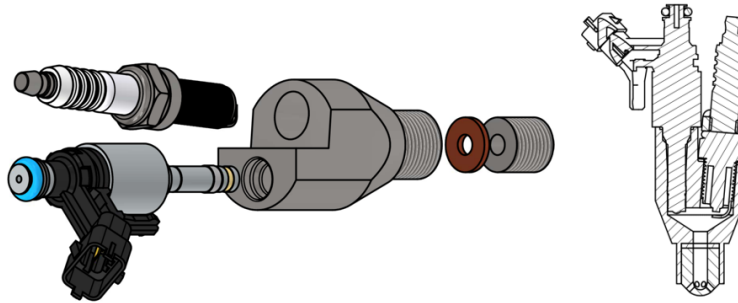


Figure 1: The design of a pre-chamber for hydrogen gas jet ignition system and its cross-section.

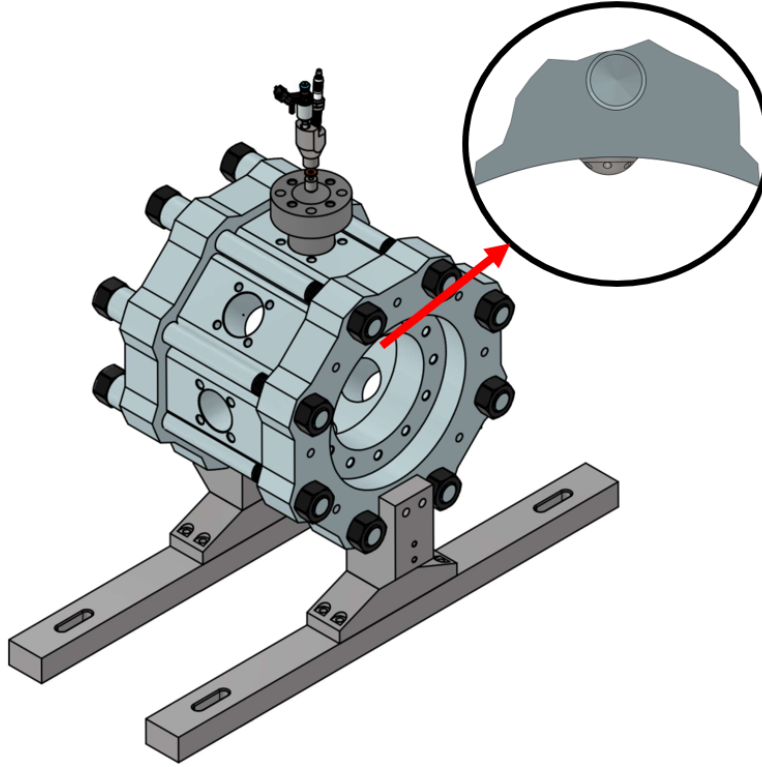


Figure 2: 3D CAD model of a constant volume combustion chamber (CVCC) with a pre-chamber on top. The nozzle location is shown from the front view of the chamber.

To conduct the experiment under engine conditions, the same pre-chamber system will be installed into an OACIC [13], the specifications of the engine are reported in Table 1. The pre-chamber has a volume of 4.2 cc which is 3.3% of the engine clearance volume. Figure 3. shows the cylinder head of the engine with the pre-chamber system mounted on the side. The optical engine allows capturing of the pre-chamber ignition event during engine operation. Furthermore, various engine parameters such as performance and emissions will be recorded and the combustion event will be captured from the optical window on the side. Figure 4. illustrates a cross section of the cylinder head and the view from the side of the cylinder.

Table 1: Engine specifications

Engine type	4-stroke, single cylinder
Bore/Stroke	130/140 mm
Displacement volume	1.85 L
Compression ratio	15.93:1
Compression ratio with pre-chamber	15.41:1

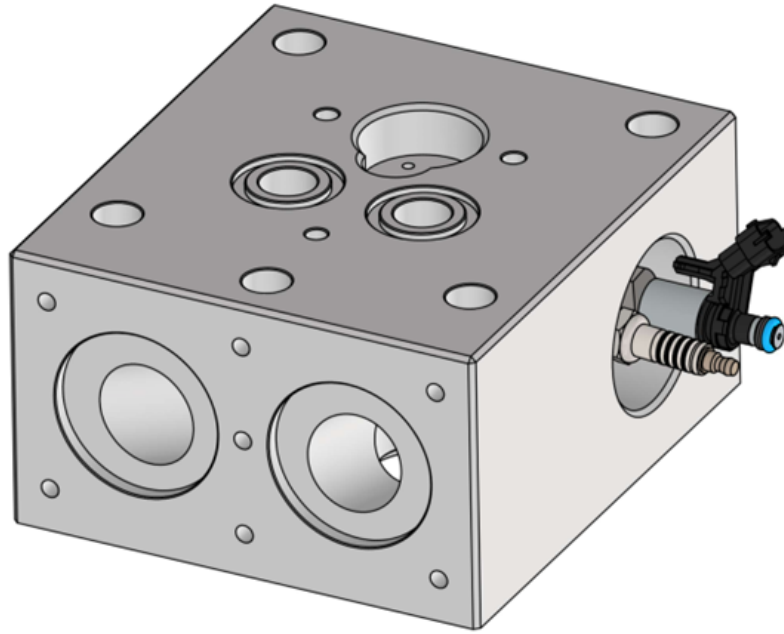


Figure 3: The cylinder head of the engine with the pre-chamber system mounted on the side.

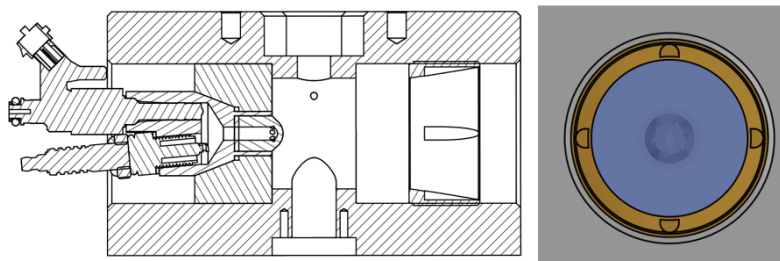


Figure 4: Cross-section of the cylinder head showing the interior view from the side of the cylinder. The pre-chamber system is visible mounted on the cylinder wall.

Expected results

This study aims to provide comprehensive results of utilising ammonia and hydrogen blends as a fuel for marine engines. The computational modelling will further improve the fundamental of the in-cylinder combustion processes and kinetics involved when using ammonia and hydrogen as the fuel. The effectiveness of hydrogen as a combustion promoter for ammonia is expected to be quantified across a range of operating conditions. The CFD simulations are anticipated to predict the spray, combustion, and emissions characteristics of the fuel blends. The experimental work will demonstrate the feasible of pre-chambers as an ignition system for ammonia-hydrogen fuels. Optical diagnostics will capture the ignition and combustion events, validating the computational models. Ultimately, the findings are expected to highlight the potential of ammonia-hydrogen blends to serve as a decarbonised, renewable fuel option for the maritime sector. The data will showcase the operational feasibility and benefits of ammonia/hydrogen mixture over conventional fossil fuels.

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